From Footage to Ethogram: An Automated Vision–Language Pipeline for Quantifying Behaviour and Affective State of a Captive Octopus

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***Abstract*—Continuous camera monitoring of captive marine animals produces far more video than caretakers can review, yet most of it is used only reactively. We present an automated vision–language pipeline that converts continuous multi-camera aquarium footage of a single *Octopus vulgaris* into a quantified behavioural and affective-state time-series. It detects when the animal is visible (a frozen CLIP encoder with a lightweight probe under aspect-preserving preprocessing), extracts short behaviour clips gated on visibility and motion, and re-prompts a large vision–language model for a structured behavioural record per clip (behaviour class, posture, activity, location, context and body colour). Aggregating $3{,}083$ analysed clips yields an interpretable activity budget, an exposure-normalised circadian profile with a late-afternoon activity peak, and a clear stimulus response in which human presence nearly doubles movement and raises a transparent arousal index. To make the system self-hosting we distil the teacher into small local models: a captioning student (Qwen3-VL-2B [7], QLoRA) running 4-bit on a laptop, and a single-class mask model auto-labelled by GroundingDINO and SAM2 and anchored by human-verified masks. On a leak-free human-labelled test (122 frames from five held-out source videos) that model reaches 0.64 mean / 0.72 median IoU with 1.1% body-area error, and its mask area doubles as the pipeline's presence gate—AUC 0.91 against human-verified empty frames from 52 recordings, and indistinguishably 0.91 against tank-glass reflections, retiring our assumption that reflections were the dominant failure mode—which works only because the student is trained with empty-mask negatives. On top of the masks we extract an *anatomical skeleton*—mantle, head and up to eight arm splines—scored on a frozen benchmark by arm-tip F1 against protrusions of the human mask (0.54), a metric adopted only after "arms per frame" proved gameable and the replacement's own ground truth proved wrong. Its occlusion-honest arm kinematics independently separate the VLM-derived behaviour classes on 146 clips from 66 source videos, aggregated to one value per (video, class) before testing (median arm-tip speed $53\,\mathrm{px\,s^{-1}}$ resting vs. 141 reaching out of water; Kruskal–Wallis $p{=}3.5\times10^{-6}$, $\varepsilon^2{=}0.22$, and still significant in scale-invariant body-lengths per second): a cross-validation of two signal paths that share no components. We frame results as arousal and behavioural state rather than emotion, measure the reliability of the very labels the analysis groups by (Cohen's $\kappa{=}0.55$ under a disjoint frame sample—moderate, and a bound on every aggregate here), report the limitations and the measured negative results honestly, and position the work as the profiling layer complementary to real-time single-behaviour response systems.**

***Index Terms*—Computer Vision; Vision–Language Models; Behaviour Analysis; Marine Animal Welfare; Model Distillation; Edge AI; Cephalopods**

I. Introduction

Aquarium and controlled-habitat monitoring increasingly relies on continuous camera coverage, but the resulting video vastly exceeds what caretakers can review. Most footage is therefore consulted only reactively, and the record of an animal's day-to-day behaviour is left unquantified. For welfare-oriented husbandry the more valuable question is longitudinal: *how does this animal spend its time, and how does it respond to its environment?*

Octopuses make this both important and hard. They are highly intelligent, behaviourally complex invertebrates whose welfare is a growing concern, yet their non-rigid bodies, rapid posture changes, active camouflage, frequent denning, and the reflections and floating debris of tank environments defeat naive visual monitoring [1]. A companion line of work from our consortium targets a single, actionable behaviour—a "protest" limb gesture used to request tank lighting—and actuates the lights; that system is reactive and behaviour-specific by design.

This paper addresses the complementary problem: *understanding the whole behavioural repertoire over time*. We present an automated pipeline that turns continuous multi-camera footage of one captive *O. vulgaris* ("Nity") into a structured behavioural and affective-state time-series, together with the small local models needed to run it cheaply and offline.

One constraint shapes the entire design. There is no large-scale, publicly available annotated dataset for octopus segmentation or detection to train on—an absence independently reported by concurrent work on octopus segmentation in natural habitats, which cites "the absence of large-scale, publicly available, annotated video datasets for octopus segmentation, which prevents standard supervised fine-tuning and evaluation" [9]. So supervised training on an existing corpus was never an option, and every label in this pipeline has to be manufactured: by foundation-model teachers run offline over our own footage, by a human wherever the measurement depends on it, or not at all. That is why the architecture is teacher→student distillation rather than fine-tuning, and why so much of what follows is about verifying labels we generated ourselves. Our contributions are:

1) An end-to-end pipeline from raw footage to a per-clip *structured behavioural record*, using a large vision–language model (VLM) as a zero-shot behavioural annotator with snap-to-list validation.
2) An aggregate behavioural analysis over a nine-day observation window (3,083 present clips, seven recording days): an interpretable activity budget, an exposure-normalised circadian profile, and a quantified stimulus response, framed as arousal / behavioural state.
3) Teacher→student distillation into deployable local models—a 4-bit captioning student that runs on a laptop, and a single-class octopus segmentation model auto-labelled by GroundingDINO+SAM2, anchored by human-verified masks, and evaluated leak-free on a held-out human-labelled test (0.64 mean / 0.72 median IoU, 1.1% body-area error — one model, one test set, with the split re-derived to prove the holdout was never trained on).
4) An anatomical skeleton and kinematics layer on the masks: per-frame mantle/head/arm graphs scored by arm-tip F1 on a frozen benchmark, with the arm-selection gates reported as an explicit precision–recall frontier rather than a single tuned point; occlusion-honest tracking; and per-arm kinematics that separate the VLM behaviour classes with video-level statistics on 146 clips from 66 recordings—a cross-validation of two unrelated signal paths.
5) A frozen benchmark suite (segmentation, skeleton, tracking; one runner, tagged results, video-level splits) and an honest account of the methodology around it, including measured negative results: a train–test leakage we caught and corrected, a headline metric retired because a genuine improvement made it worse, a metric ground truth we had to validate and fix, a null result retiring our own assumption about the pipeline's dominant failure mode, and several plausible improvements that failed under the fixed benchmark.

II. Related Work

Cephalopod behaviour and welfare. Interest in octopus cognition and welfare has grown alongside their inclusion in animal-welfare frameworks. Prior consortium work introduced a framework for decoding inter-species communication with octopodes and a real-time detector for a light-request behaviour that actuates the habitat [1]; our work is the profiling counterpart to that response system.

Foundation models as annotators. General-purpose VLMs describe and classify images zero-shot, and open-vocabulary detectors and promptable segmenters (GroundingDINO [3], SAM/SAM2 [4]) label without task-specific supervision. We use these as *teachers*, distilling their labels into small students [5]. CLIP [2] provides the frozen features for our presence detector; the deployed segmenter uses an LR-ASPP/MobileNetV3 head [6]. Off the shelf, however, these models were not usable as detectors on our footage; Sec. III-D quantifies where they failed.

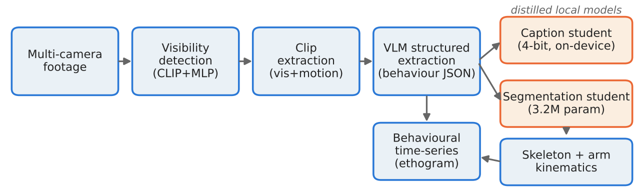

Fig. 1. Footage-to-ethogram pipeline (the four stages of Sec. III): visibility detection → clip extraction → VLM structured behavioural extraction → distillation into local caption and segmentation students.

Octopus segmentation in the field. The closest work to ours is HideAndSeg [9], a concurrent annotation-light segmentation pipeline: a human annotates initial frames to prompt SAM2, the resulting masks train a YOLOv11 detector, and that detector then auto-prompts SAM2 over the remaining footage. On 148 videos of juvenile *Octopus insularis* (366,514 usable frames) recorded at three Brazilian coastal sites, it reports DICE 0.9677 and IoU 0.9383.

Those figures are not comparable with the 0.641 IoU we report, and we would rather say so than let the contrast be drawn for us. (i) *The reference differs.* Their ground truth is manual segmentation of the initial frame of each video—essentially the frame used to prompt SAM2—while ours is 122 human-drawn masks sampled across five source videos held out of *all* training. (ii) *The imaging conditions differ*: natural-habitat colour video against dim aquarium footage with infrared cameras, tank-glass reflections and clutter. (iii) *The subject differs*: juvenile *O. insularis* against an adult *O. vulgaris*. (iv) *The deployment target differs*, which is our actual distinction: HideAndSeg runs SAM2 at inference, whereas we distil to a 3.2 M-parameter student that runs locally with no GPU and no API. The two lines are complementary rather than competing—they solve automated *prompting* of a large model in the field, we solve *distillation to a deployable model* and the behavioural layer built on it. That both projects had to manufacture their own labels is the same finding reported twice.

III. System and Methods

The tank is observed by a multi-camera rig (6 IP cameras, 4K, infrared and colour night vision) with a manually selected region of interest around the den per camera; the same installation supports the companion response system.

A. Visibility Detection

A frozen CLIP ViT-B/32 encoder feeds a small multi-layer-perceptron probe (512→256→64→2) classifying *octopus-visible* vs. *hidden*. The single most important design choice was preprocessing: CLIP's default centre-crop discards 33–44% of a 16:9 frame, and *aspect-ratio mismatch*—not backbone capacity—was the root cause of poor field performance. Aspect-preserving letterbox padding plus ∼1.7k mined reflection/infrared-noise hard negatives gave a probe reaching 96.8% on our internal detection test set.

B. Clip Extraction

Per second of source video we compute the probe's visibility probability and an absolute changed-pixel motion fraction (with the burned-in timestamp region masked so the ticking clock is not counted as motion), and keep a non-overlapping $20\,\mathrm{s}$ window when the octopus is visible for a majority of it and mean motion exceeds a threshold. The *absolute* fraction is deliberate: per-video normalised motion promotes static scenes (a flickering lamp) to "motion".

C. Structured Behavioural Extraction

Rather than treating a caption as the end product, we treat it as one field of a *structured behavioural record*. Per clip we select the clearest frames (contrast-enhanced for the dim infrared footage, scored by the visibility probe) and prompt a large VLM (Qwen3-VL-235B [7], via API) for a JSON record: {present, behaviour (7-class ethogram), posture, activity level, location, context, body colour, colour/texture change, confidence}. Each field is validated by snapping to an allowed value list; colour fields are gated per clip, so infrared/greyscale clips (detected by low BGR channel divergence) are told to leave colour *uncertain*. The run over 3,205 clips cost $\approx$ \$2.22 ($\sim$ \$0.0006/clip). One field of the schema is dead and we report it as such: *colour/texture change* was never once asserted across the 99 colour clips of the reliability sample, so it carries no signal and is a candidate for redesign. The visibility probe is the weaker of our two presence signals; Sec. V measures it head-to-head against the mask-area gate rather than by anecdote.

D. Distillation to Local Models

The expensive foundation models are used only as *offline teachers*; nothing large is deployed. Their outputs become training data for small students, so the running system is self-hosting and needs no GPU server or paid API. We apply the recipe twice—VLM captions into a compact captioning model, GroundingDINO+SAM2 masks into a single-class segmentation model (Sec. V)—each kept as small as the task allows so continuous in-situ inference is feasible on commodity hardware.

Why distil rather than prompt. Distillation is not only a cost argument: we measured the off-the-shelf alternatives on this footage and none of them worked as a detector. Asked to verify 232 mined hard-negative frames, an open-vocabulary detector (OWLv2 [8]) reported an octopus in 231 of them at a 0.10 threshold and its scores never separated the two classes—no operating point existed, so it was useless as an auto-filter. Human review of the same frames found that 166 genuinely contained the animal and 66 did not, a distinction the detector had not made. Zero-shot CLIP scoring for presence was likewise abandoned as unreliable; what works is a *trained* probe on frozen CLIP features (Sec. III-A), and even that is the weaker of our two presence signals. The promptable segmentation teacher is usable, but only when gated by seed confidence and propagated through time, and it fails outright on one of our cameras (Sec. V). Read together, these say the same thing: the foundation models here are *gated, temporally propagated offline annotators*, not drop-in detectors—which is exactly the case for distilling them into a student rather than prompting them at inference time.

E. Diverse-Footage Harvesting

Early models were trained on footage from a single week, the binding generalisation limit. We therefore built a streaming harvester that crawls the archive and *probes* a handful of frames per candidate video before committing to a full scan, skipping empty "den" videos cheaply. Because the server imposes a low aggregate bandwidth cap (measured $\sim$5 MB/s, parallel connections yielding only $1.6\times$), it runs as a single CPU worker that streams rather than bulk-downloads, keeping a coverage ledger so skipped footage can be revisited—expanding coverage from 7 to >100 distinct days.

F. Evaluation Protocol: a Frozen Benchmark Suite

A pipeline this long makes it easy to accumulate numbers each measured on a slightly different set, so we froze three benchmarks behind a single runner writing tagged results, making every improvement claim a before/after on the same data. *SEG-TEST*: 122 human-labelled mask frames plus 19 empty-mask negatives, from five source videos excluded from *every* training source; mask IoU, body-area error, presence AUC. *SKEL-50*: 50 frozen frames over 20 source videos, each with image, human mask and model mask; arm-tip F1. *TRACK-10*: 10 frozen clips spanning all six behaviour classes, three cameras and three dates; teleport rate, occluded fraction, coverage. Two rules hold throughout: splits are by *source video*, never by frame—frames from one recording are near-duplicates and we shipped that bug once (Sec. V)—and the frozen lists are never regenerated to suit a result (Table I).

We also verify "held out" rather than assert it. All five SEG-TEST videos appear in the deployed model's *dataset* manifest, which lists every labelled pair, but a forced-holdout flag keeps them out of training: re-deriving the split from the manifest, source filter and seeded shuffle reproduces the logged partition exactly (3,450 train / 1,693 validation frames), with no holdout video in training. The audit also exposed—and we fixed—a logging bug printing wrong video counts for a correct split, which is what a leak looks like at a glance.

IV. Behavioural and Affective-State Analysis

Aggregating the 3,083 present clips (joined to absolute clock time) yields an interpretable behavioural profile. These clips come from *seven recording days spanning nine calendar days* (20–28 February 2026), so the profile below characterises one observation window rather than a season: the circadian curve in particular is an average over seven days, and we make no claim about seasonal or long-term drift. Extending this analysis across the harvested $\sim$ 209-day corpus is the immediate next step (Sec. VII). We report **arousal and behavioural state**, not emotion; the transparent arousal index is $0.6 \cdot \text{activity} + 0.4 \cdot \text{posture-spread}$, with static body colour a secondary channel on colour cameras only.

TABLE I

FROZEN BENCHMARK SUITE, DEPLOYED CONFIGURATION (OCTO_SEG_THIN768_LRASPP.PT). ALL SPLITS ARE BY SOURCE VIDEO; THE SEGMENTATION TEST VIDEOS ARE EXCLUDED FROM EVERY TRAINING SOURCE. MASK IOU COMES FROM THE SAME MODEL AND TEST SET AS THE PRESENCE ROW, BUT THAT ROW'S 19 NEGATIVES COME FROM ONLY TWO RECORDINGS; IT IS RETAINED FOR CONTINUITY AND IS SUPERSEDED BY THE MULTI-RECORDING HUMAN-VERIFIED NEGATIVE SETS REPORTED IN THE TEXT. TIP-F1 SCORES ARM RECOVERY AGAINST THE HUMAN MASK'S PROTRUSIONS, SO BOTH MISSED AND SPURIOUS ARMS ARE PENALISED; ARM COUNT IS DESCRIPTIVE ONLY.

Metric	Value
SEG-TEST — 122 human-mask frames, 5 held-out videos	
mask IoU (mean / median)	0.641 / 0.719
body-area error	1.05%
presence AUC (vs 19 empty, 2 recordings)	0.794
SKEL-50 — 50 frozen frames	
arm-tip F1	0.539
tip precision / recall	0.722 / 0.502
arms per frame (descriptive)	3.68

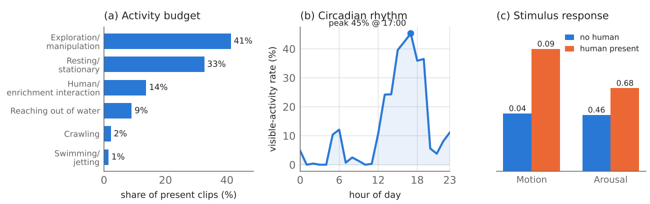

Fig. 2. Aggregate behaviour from the structured records: (a) activity budget; (b) exposure-normalised circadian visible-activity rate; (c) motion and arousal with vs. without human presence.

Activity budget. Exploration/manipulation 41%, resting 33%, human/enrichment interaction 14%, reaching out of water 9%, crawling 2%, swimming/jetting 1% (Fig. 2a).

Circadian structure. Normalising present-clip counts by all extracted windows per hour (an exposure correction) reveals a strong diel pattern: visible activity is ~1–5% overnight and rises to a ~45% peak at 17:00 with a 13:00–19:00 plateau, plus a dawn bump at 05:00–06:00.

Stimulus response. With vs. without a human present, mean motion nearly doubles ($0.045 \rightarrow 0.095$) and the arousal index rises ($0.46 \rightarrow 0.68$). On colour cameras the dark red-brown body colour is most common at baseline (~16%) and less so during human interaction (~6%). We regard such *contrasts* as the robust findings; absolute levels depend on the upstream presence gate and will shift as the detector improves.

V. LOCAL MODELS

A. Captioning Student

We distil the teacher's one-sentence descriptions into Qwen3-VL-2B [7] using QLoRA (rank 16). On a 50-clip held-out split, embedding similarity to the teacher improves from 0.702 (base) to 0.834 and ROUGE-L from 0.269 to 0.455. Quantised to 4-bit with MLX, the student is ~1.7 GB and captions a clip in ~3 s on a 16 GB laptop with no GPU.

B. Segmentation Student

Following Sec. III-D we target the *smallest* model that yields usable masks: one foreground class at low resolution, with GroundingDINO (box) + SAM2 (mask) as an offline auto-labeller, never deployed. The most-confident frame seeds SAM2 video propagation, and largest-blob plus area-continuity cleanup remove transient errors.

That recipe is a correction, not a refinement. Prompting the teacher independently per frame over-segments in two systematic directions, both measurable: on the infrared camera SAM2 attaches to bright metal tools, and seeding a single propagated track cuts mask area from 11.8% to 6.5% of the frame; on colour cameras the mask escapes the loosely fitted box into the background, 15.5% to 5.8%. Clips already clean show no regression. Seed confidence supplies the rest: on the tank-glass reflection camera GroundingDINO's best frame peaks at ≈ 0.50 against 0.74–0.89 on real animals, so the camera is rejected wholesale rather than quietly mislabelled. The limit of the recipe is worth stating with its success, because it is the same mechanism: temporal propagation removes errors that are *transient* across a clip, and cannot touch errors that are *consistent*—a reflection is present in every frame—which is why confidence gating is needed alongside it and not instead of it. Nor does the teacher transfer across sensors: on greyscale infrared clips GroundingDINO is so rarely confident enough to seed that only 13% of clips (87 of 653) pass the gate, and the masks that do survive over-segment onto bright fixtures (median area 8.5% against 2.9% on colour), leaving our largest deployment camera unlabelled (Sec. VII). Auto-labelling produced 4,412 (image, mask) pairs, from which we distilled students across a size sweep (a compact U-Net at several widths, an LR-ASPP head); we retain the smallest clearing our quality bar, an LR-ASPP student of 3.2 M parameters.

Training configuration. 768^2 input, 60 epochs, batch 8, Adam at 3×10^{-4} (cosine), best-validation-IoU checkpoint; loss = focal Tversky (α=0.2, β=0.8, so false negatives—missed thin arms—dominate) + 0.5 BCE; strong augmentation applied to image and mask in lock-step (flip, affine, ±25% brightness–contrast, sensor noise); 5,143 pairs from 183 source videos, split by source video. Code, frozen benchmark definitions and evaluation scripts will be released with the paper.

Mask area as the presence gate. Mask area doubles as the pipeline's presence signal, and works only because the student sees empty-mask negatives: a positives-only student was a coin flip (AUC 0.50). Measuring it required negatives the detector had no hand in choosing, so we sampled frames at uniform random timestamps from whole source videos—not from extracted clips, which exist only where the visibility probe fired—excluded every recording session used to train either model, and had every frame *human*-verified before scoring; that step is not a formality, since 9 of the 105 frames our own vision-model pass called empty contained the animal. The deployed model reaches presence AUC 0.907 on 97 verified empty frames from 52 recordings (Table II), *superseding* the

TABLE II

PRESENCE GATE ON **HUMAN-VERIFIED** NEGATIVES (DEPLOYED SEGMENTER, THRESHOLD 0.5; CIS CLUSTER-BOOTSTRAPPED BY SOURCE VIDEO). NEGATIVES ARE DRAWN AT UNIFORM RANDOM TIMESTAMPS FROM WHOLE SOURCE VIDEOS, SO EACH SET IS INDEPENDENT OF THE GATE BEING TESTED. TOP: THE TWO FAILURE MODES, MEASURED SEPARATELY AND NEVER POOLED. BOTTOM: BOTH GATES ON ONE IDENTICAL SUBSET OF THE REFLECTION FRAMES, THE PROBE'S TRAINING SESSIONS REMOVED FROM BOTH ARMS (PAIRED ΔAUC $+0.134$, $[+0.024, +0.308]$); THAT SUBSET COMES FROM EXTRACTOR-SELECTED CLIPS, SO THE GAP IS AN UPPER BOUND. FP IS THE FALSE-POSITIVE RATE AT PRESENT-RECALL 0.90.

Negatives / gate	fr / vid	AUC [CI95]	FP
Empty frames	97 / 52	0.907 [0.826,0.958]	0.186
Reflections	36 / 29	0.906 [0.806,0.956]	0.278
Head-to-head, identical frames			
mask area (zero-shot)	30 / 24	**0.913** [0.821,0.962]	0.267
CLIP probe (in-domain)	30 / 24	0.799 [0.598,0.882]	0.700

0.794 of Table I from 19 frames in two recordings—an artefact of one unusually hard recording rather than a property of the model.

A null result: reflections are not the harder case. Tank-glass reflections, on which the visibility probe fires confidently, were assumed to be the pipeline's dominant false-positive source. Against 36 verified reflection frames from 29 recordings the gate scores 0.906, indistinguishable from the empty-frame 0.907. We drop the reflection framing and assert no ordering in either direction: on this evidence the two failure modes are equally hard. At the *deployed* operating point (area ≥ 0.01) 15.5% of verified empty frames and 22.2% of reflection frames still fire, so the gate attenuates rather than removes false positives.

Head-to-head: zero-shot mask area vs. the in-domain probe. Scored on one identical verified set with the probe's training sessions removed from *both* arms, mask area beats the probe by a paired ΔAUC of $+0.134$, CI95 $[+0.024, +0.308]$, excluding zero—and the asymmetry sharpens it: the probe is *in-domain* here, trained on 1,519 frames from this camera, while the segmenter is *zero-shot*. Two caveats travel with it: the interval is wide (24 recordings), so the claim is the *ordering* and not the effect size; and these frames come from extractor-selected clips, enriched for the probe's own false positives, making $+0.134$ an upper bound.

Raising mask quality: what worked and what did not. Early students plateaued near 0.47 IoU across model sizes, architectures and augmentation. (i) *Video diversity* (sources expanded from 62 to 176 distinct videos via the harvest of Sec. III-E) helped modestly but robustly ($0.468 \to 0.494$ on a harder, diverse-date validation). (ii) *Cleaner teacher labels did not help*: re-labelling with larger teachers (GroundingDINO-base, SAM2-large) left held-out IoU flat ($0.494 \to 0.506$); the earlier signal suggesting label noise was the ceiling turned out to be *train–test leakage* between two label sources, caught, corrected with a hard video-level holdout, and re-measured. (iii) *Human-verified masks*: a click-to-SAM2 tool (one click seeds SAM2, the human corrects) produced 412 positive and 87 empty-mask labels; human labels alone underperformed the blend (0.505 vs. 0.608)—at this scale volume beats purity—but they give the project its first human-verified held-out test set. (iv) The decisive levers came from inspecting failures there: the octopus occupies a median 2.5% of the frame, so at the original 256^2 input its arms are 1–2 pixels wide, and the symmetric Dice+BCE loss did not penalise under-segmentation. Training at 512^2 with a focal Tversky loss ($\beta > \alpha$, penalising false negatives) lifts 0.466 to 0.608 mean / 0.666 median IoU; 768^2 with β=0.8, improving thin-structure recall, gives the deployed model: 0.641 **mean /** 0.719 **median IoU**, 1.05% body-area error on the same 122 frames (Table I). It also scores above the 512^2 variant on presence (0.794 vs. 0.718 AUC)—indicative only, as both share the same 19 single-recording-dominated negatives—so one checkpoint serves the presence gate and the skeleton of Sec. VI. The residual IoU gap is dominated by thin arm boundaries; the *area* signal the behavioural analysis consumes is accurate to $\sim 1\%$.

A negative that needed a control: test-time temporal fusion. Because the model fails by mis-*locating* a correctly-sized blob, the obvious remedy is temporal evidence—and the deployed skeleton path already thresholds an exponentially-smoothed map, while every IoU above is single-frame. We fused each frame with its neighbours at $t \pm 1, \pm 2$ (per-pixel median) three ways: exponential smoothing, DIS-optical-flow warping before the median, and—as a control—the same median with *no* warping. A fixed threshold would be unfair, since a median over neighbours systematically shrinks the probability map, so we swept it and scored every variant at its own best operating point. *The negative holds*: the best arm (exponential smoothing) reaches 0.5866 IoU against 0.6552 single-frame (-0.069). Test-time fusion does not repair mislocalisation; it blurs boundaries on a fast-deforming animal. The control then changed our reading of *why* fused masks gate presence better: unwarped averaging beats optical flow on *both* metrics (IoU 0.5527 vs. 0.5109; presence AUC 0.9629 vs. 0.9521), so averaging supplies the benefit and motion compensation subtracts from it—dense flow is unreliable on a low-texture, deforming body in dim infrared. Without the control we would have published "optical-flow fusion improves the presence gate": true in isolation, wrong about the mechanism. We do not headline the fusion presence gain: it rests on the same 19 two-recording empty frames and has not been re-measured on the verified negative sets above.

VI. ANATOMICAL SKELETON AND ARM KINEMATICS

The masks enable a more structured representation: a per-frame *anatomical skeleton*—mantle centre, head node and up to eight arm splines with base/mid/tip landmarks—from which per-arm kinematics (tip speed, posture spread) are derived. Extraction is classical: topology-preserving thinning, Dijkstra arm-path selection from an anatomical root, and mask-constrained splines.

A frozen benchmark and measured fixes. All skeleton work is evaluated on SKEL-50 (Sec. III-F). The headline met-

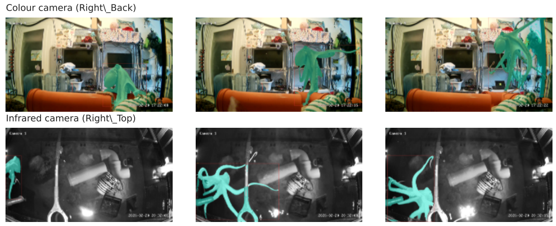

Fig. 3. Auto-labelling teacher (GroundingDINO+SAM2) masks tracked across a clip on a colour camera (top) and an infrared camera (bottom). SAM2 video propagation with largest-blob and area-continuity cleanup yields the temporally consistent masks used to train the student.

ric is *arm-tip F1* against the protrusions of the *human* mask, matched greedily one-to-one within 5% of the image diagonal, so a duplicate arm cannot match the same protrusion twice and both missed and spurious arms are penalised. Decomposing the loss per stage (silhouette $\rightarrow$ thinning $\rightarrow$ selection) located the errors and produced the fixes we ship: relaxed arm-selection floors that had discarded short, curled arms; mask preparation whose blur threshold erased thin arms; an anatomically constrained head; the higher-resolution student of Sec. V; and, offline, SAM2 refinement of the student's mask (the student *locates*, SAM2 *sharpens*). The head rule—a neck constriction between mantle and arm crown—replaced a widest-second-blob heuristic plausible in only 9% of frames; it reached 96% on introduction and reads 80% under the current stricter gates, but "plausible" is a loose criterion that we are replacing with pixel error against human clicks on the eyes. Two plausible improvements measured as nulls: hysteresis skeletonisation of the segmentation *probability field* (the missing arms are not sub-threshold—the student genuinely does not see them) and a zoom-in second segmentation pass (crops are out-of-distribution). On the ceiling: the same protrusion detector on the *human* masks yields a mean of 5.7 (median 6, max 8) protrusions per frame—a 2-D silhouette hides arms folded under the body—so even perfect recall here is recall against a subset of the eight arms.

Choosing an operating point, not a number. Arm selection carries two "anti-mess" gates—unique-suffix and tip-thinness—added after inspection showed duplicate late-forking arms and stubs ending inside the body. They are a straight precision-for-recall trade, so we report the whole frontier (Table III), not just the shipped point. We ship an F1-suboptimal setting because downstream kinematics consume arm *identities* over time, where a duplicated arm is worse than a missing one.

Two lessons about metrics. First, *arm count is not a score*: when the gates removed roughly 1.3 duplicate or tangled arms per frame—a visibly cleaner output—the count we tracked fell $4.80 \rightarrow 3.48$, a 27% "regression" produced by a genuine improvement. Precision alone is equally gameable (emit one obvious arm), hence F1 against human-mask protrusions. Second, *the replacement metric's own ground truth was wrong*: at default sensitivity our protrusion detector counted contour bumps as separate arms—mean 9.7, up to 14 per human

TABLE III

ARM-SELECTION GATE FRONTIER ON SKEL-50 (50 FROZEN FRAMES, CORRECTED TIP GROUND TRUTH). GATES ARE (UNIQUE-SUFFIX SCALE, UNIQUE-SUFFIX FRACTION, TIP-THINNESS RATIO); P/R ARE AGAINST HUMAN-MASK PROTRUSIONS AND ARMS PER FRAME IS DESCRIPTIVE ONLY. *Caveat:* THE DUPLICATE RATE USES THE SAME CRITERION THE GATES ENFORCE, SO GATED ROWS ARE 0 BY CONSTRUCTION.

Gate setting	P	R	F1	Dup.	Arms
Gates off	0.659	**0.562**	**0.565**	0.076	4.64
$(1.0, 0.15, 0.85)$	0.673	0.535	0.550	0.038	4.28
$(1.5, 0.20, 1.00)$ shipped	0.722	0.502	0.539	0.000	3.68
$(2.0, 0.30, 0.55)$ 1st attempt	**0.760**	0.468	0.520	0.000	3.24

Fig. 4. Qualitative example from the frozen benchmark (arm splines coloured, tips marked), shipped post-gate configuration. Left: skeleton on the segmentation student's mask (3 arms). Right: the same frame after offline SAM2 refinement, which sharpens the silhouette and recovers 5 arms; the rest are folded against the body, not separable in a 2-D silhouette.

mask—so the eight-arm cap bound in 80% of frames and recall was scored against partly fictional arms. Requiring peaks $\geq$10% of the contour apart moved the shipped configuration's tip-F1 from 0.441 to 0.539 (precision $0.685 \rightarrow 0.722$, recall $0.380 \rightarrow 0.502$). The error was concentrated in recall, so every pre-fix F1 is incomparable with post-fix ones and every ranking had to be re-measured (Table III). Validate a new metric's *own* ground truth before trusting rankings it produces.

Occlusion-honest tracking and kinematics. Across a clip, per-frame skeletons are linked by seeding from the best-resolved frame and propagating bidirectionally (median arms maintained per clip rise from 2.2 to 5.8), with a per-node optical-flow motion prior. Every node carries a state—*detected*, *fitted* or *occluded*—and motion is computed *only* from evidence-backed samples: instrumenting this revealed that 42% of arm samples in naive tracking were evidence-free holds whose apparent smoothness masked fabricated stillness. A global tracklet-association alternative measured worse and is reported as a negative.

Cross-validation of the two signal paths. Merging per-clip kinematics into the structured records gives an independent check on the VLM labels, tested on a behaviour-stratified sample of 146 clips from 66 source videos ($\sim$25 clips per class, at most two per video) and aggregated to one value per (video, class) before any test, since clips from one recording are not independent. State-gated arm-tip speed separates the classes in the direction ethology expects (Fig. 5): median $53\,\mathrm{px\,s^{-1}}$ for resting, rising through human interaction, crawling, swimming and exploration to 141 for reaching out of water (Kruskal–Wallis H=33.2, p=3.5×10^{-6}, ε^2=0.22 over 132 video–class units; all five resting-vs-class contrasts significant after

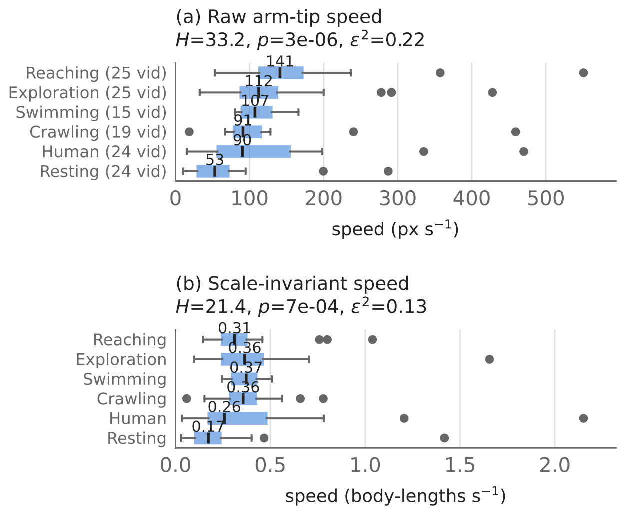

Fig. 5. State-gated arm-tip speed by VLM-assigned behaviour class (146 clips / 66 source videos, ≤ 2 per video). Each box summarises the per-*video* medians—the independent unit, since clip-level pooling would be pseudo-replication; dots are videos beyond the whiskers. (b) divides speed by arm spread, removing the apparent-size confound; both panels keep the class order of (a), so the rank change is visible: reaching is fastest in raw pixels but third in body-lengths per second, being performed with an extended body.

Holm correction). Raw pixel speed confounds motion with apparent size, so every test is repeated on speed divided by arm spread—body-lengths per second—and the separation survives (H=21.4, p=6.8×10^{-4}, ε^2=0.13, all five contrasts still significant). The normalised view also corrects the raw one: reaching is fastest in pixels but only third in body-lengths per second, being performed with an extended body, while swimming and crawling rise. Since the two pathways share no components beyond the raw video, their agreement indicates both track real behaviour rather than each other's artefacts—though it shows the kinematics follow the labels, not that the labels are correct. The labels are in fact only moderately self-consistent (Cohen's κ=0.55, below), which cuts in this result's favour: label noise unrelated to motion dilutes a group-difference test toward the null, so the separation found here is likely conservative. An earlier n=40-clip version, pooled at clip level, gave the same ordering with more extreme endpoints (63 to 159 $\mathrm{px\,s^{-1}}$); the powered estimate is the more conservative one. Speeds are camera-relative pixels; metric calibration is future work.

Relevance to the OCEANS community. The work targets intelligent sensing and longitudinal behavioural analysis for controlled marine habitats, and emphasises *edge deployability*: the students run offline on commodity hardware, so in-situ monitoring needs no cloud. Where the companion system acts on one behaviour in real time, this pipeline supplies the baseline against which such events are interpreted.

VII. Limitations and Ongoing Work

We are deliberate about what is not yet established.

- **Human-verified test sets exist only for segmentation** (499 human masks; five source videos excluded from *all* training sources). Detection and captioning stay on internal splits, so we make no comparative claims there.
- **The behaviour labels are only moderately self-consistent, and every aggregate result is grouped by them.** Re-annotating a frozen sample of 250 clips from 140 recordings with a *disjoint* set of input frames changes roughly a third of the behaviour labels: raw agreement 0.652, Cohen's κ=0.552, CI95 $[0.472, 0.624]$—"moderate". This bounds the activity budget, circadian profile, stimulus response and kinematics cross-validation alike. It measures *consistency under frame sampling*, not accuracy: a perfectly consistent extractor can be consistently wrong, and validation against a human ethologist remains the larger open item. Noise of this kind, if independent of motion, attenuates group differences toward the null, so the kinematics separation (Sec. VI) is likely understated rather than inflated—an argument that fails if fast clips are preferentially mislabelled, which we cannot yet test. The rare classes are least stable (swimming/jetting 43% reproduced, n=7; reaching out of water 48%, n=21) and are also the classes with the fewest recordings in the kinematics test, so per-class claims about them deserve extra caution.
- **Leakage is easy and we hit it once.** Two label sources sharing source videos silently placed test videos in training; an apparent $0.49 \to 0.70$ improvement evaporated under a hard video-level holdout. All results here post-date that correction and the current holdout was re-verified (Sec. III-F).
- **The presence gate is now powered, but not unbiased everywhere.** The frozen suite's 19 empty-tank negatives (two recordings) are superseded by the human-verified multi-recording sets of Table II; the reflection subset used for the head-to-head, however, is still drawn from extractor-selected clips, so that comparison's effect size is an upper bound. Absolute behavioural rates depend on this gate; we report contrasts, which are robust to a constant gate bias.
- **Mask errors are localisation errors, and the cheap temporal fix fails.** Body area is accurate to $\sim 1\%$ while IoU is 0.64: the mask is the right size in the wrong place. Test-time fusion does not repair this even at each variant's best threshold; a temporally *trained* student is untested.
- **Context label is coarse.** "Enrichment interaction" fires whenever a permanent tank object is visible ($\sim$66% of clips); the clean stimulus contrast is human-present vs. absent.
- **Skeleton and kinematics.** Arm recovery is silhouette-limited: even human masks expose only 5.7 arm protrusions per frame and the model's masks cost more, so tip recall stands at 0.50. Kinematics cover a 146-clip stratified sample, with swimming/jetting on the fewest recordings (15), and infrared cameras are not covered by the colour-trained segmenter. A static-feature behaviour classifier failed (below the majority baseline), confirming that behaviour classification needs temporal features rather than more labels—which

the kinematics layer supplies.

Ongoing work: extending the kinematics to the full corpus and validating the behaviour labels against a human ethologist; distilling the structured extractor into the local 2-B student (JSON targets rather than a sentence); and, to pass the silhouette ceiling, a learned keypoint model on RGB appearance rather than masks.

VIII. Conclusion

We presented an automated vision–language pipeline that converts continuous octopus footage into a quantified behavioural and affective-state time-series and distils its large teachers into local models that run offline. It surfaces an interpretable activity budget, a circadian rhythm and a measurable stimulus response; the segmentation student—evaluated leak-free on a human-verified test—supplies the presence signal and the masks for an anatomical skeleton whose occlusion-honest kinematics separate the language-model behaviour classes under video-level statistics. Reported with its negative results and leakage lessons, it offers a practical, welfare-oriented layer for longitudinal behavioural understanding in controlled marine habitats.

Ethical Statement and Data Availability

The work was conducted in a private aquarium, with only sub-threshold interactions w.r.t. EC directive 2010/63/EU of several caretakers with a single *O. vulgaris* (supplier: Flying Sharks, Portugal). Husbandry and enrichment followed best practice under FELASA guidelines. Video data are available from the last author.

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